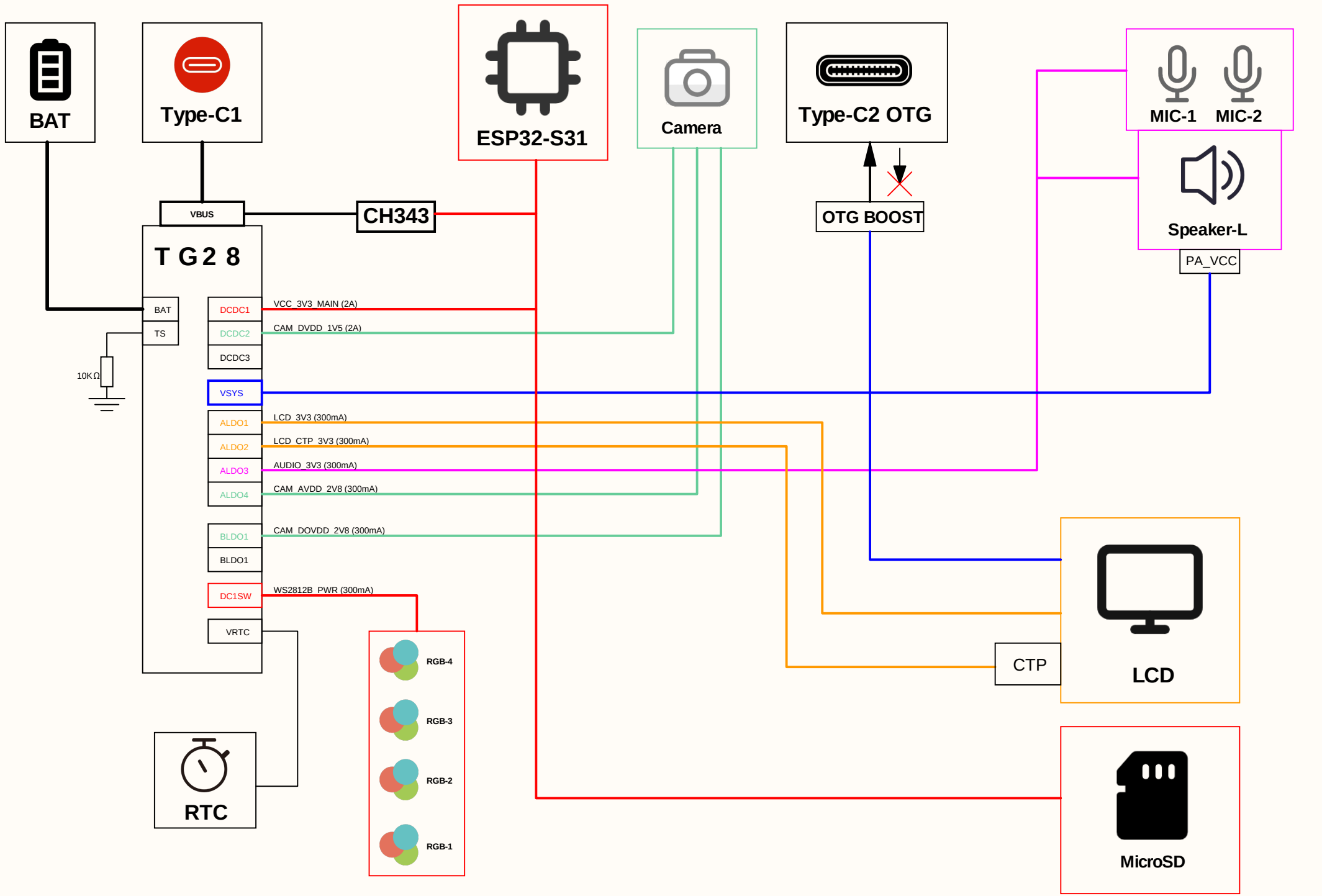
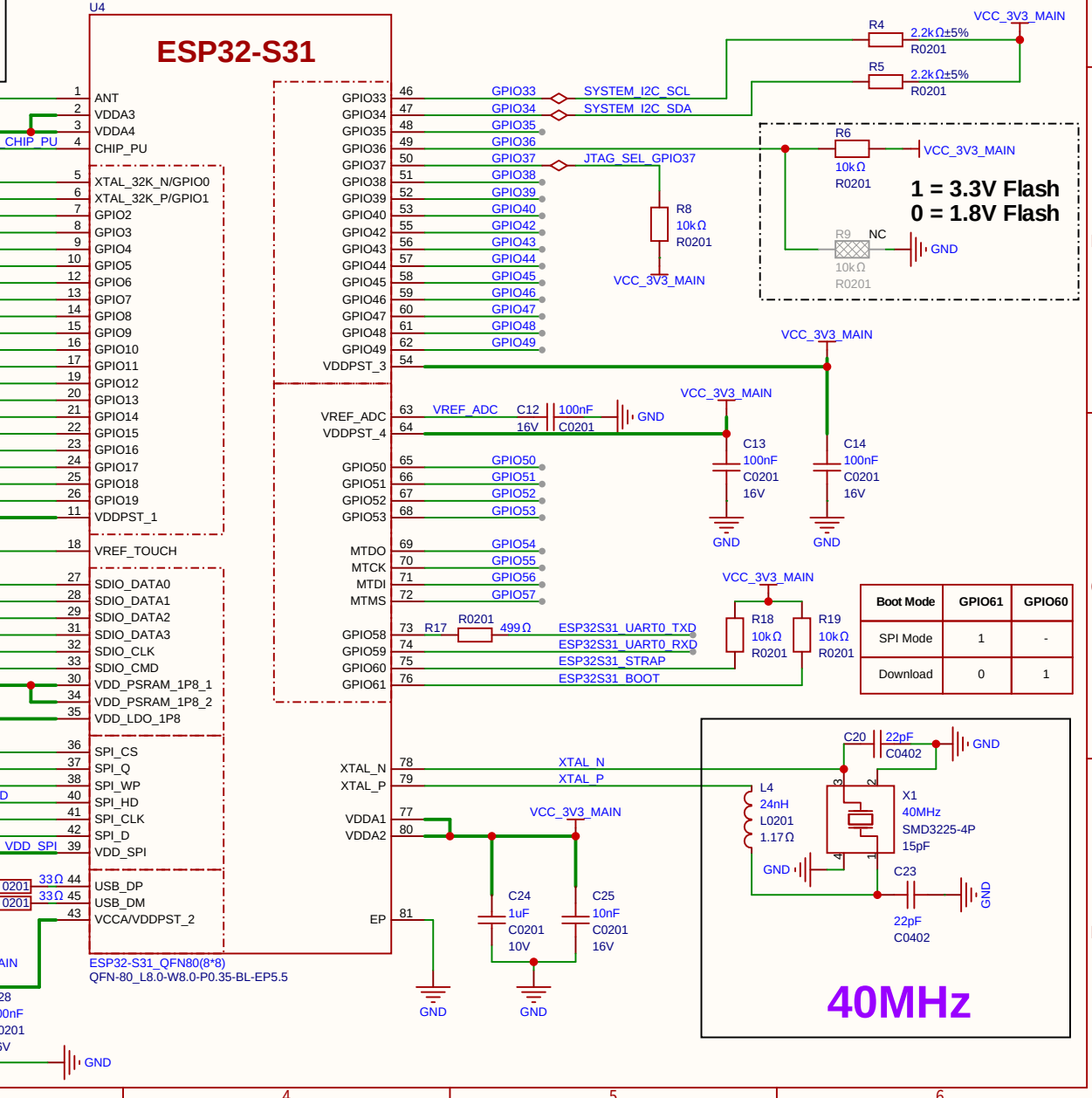
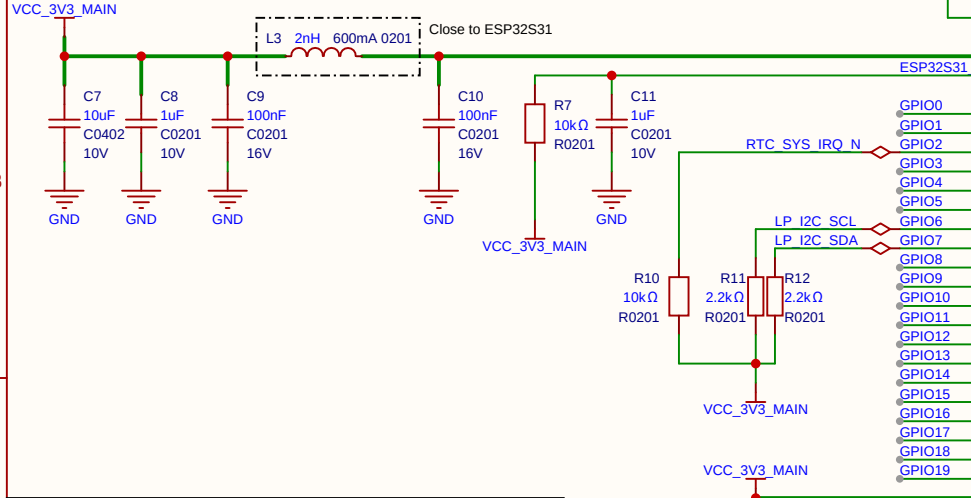
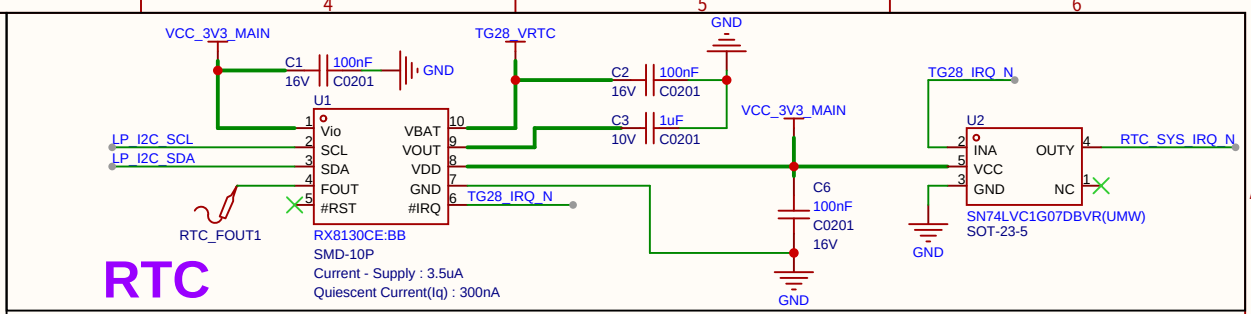
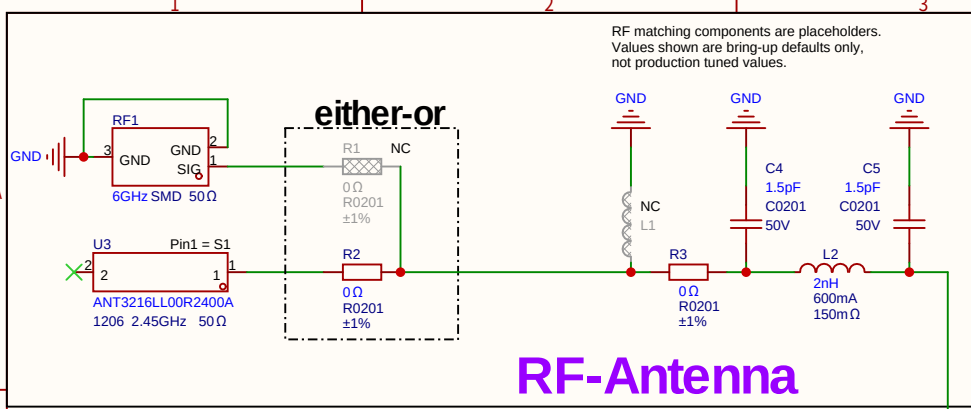
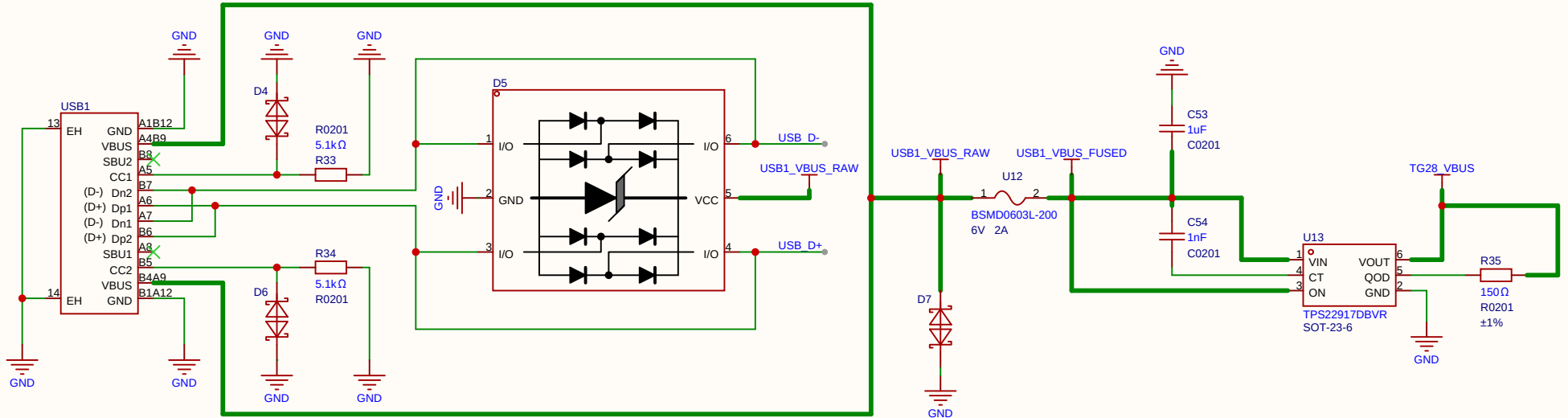


	1	2	3	4	5	6
A						
B						
C						
D						

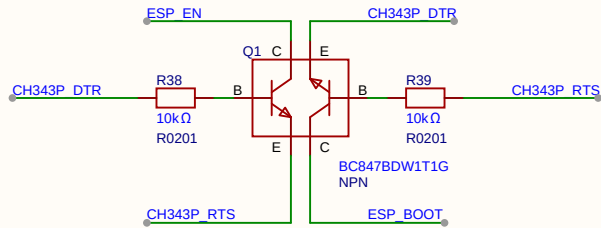
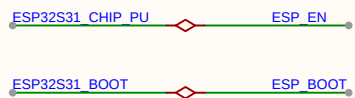




USB Type-C

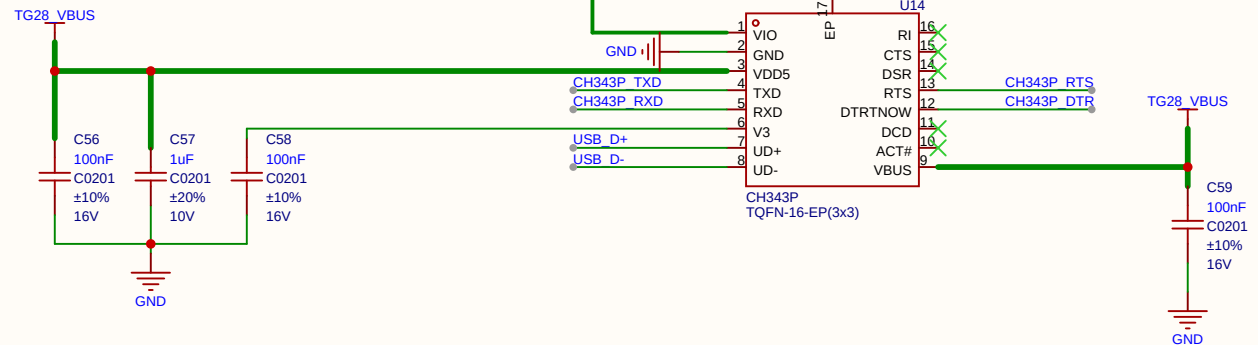
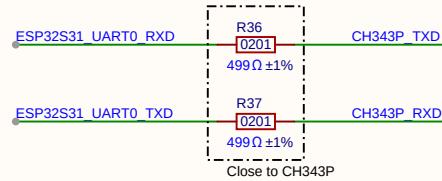


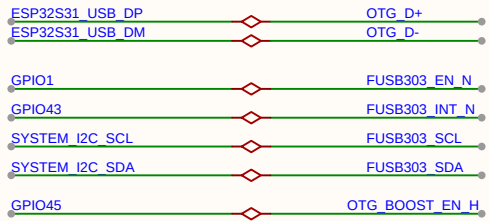
Auto Download



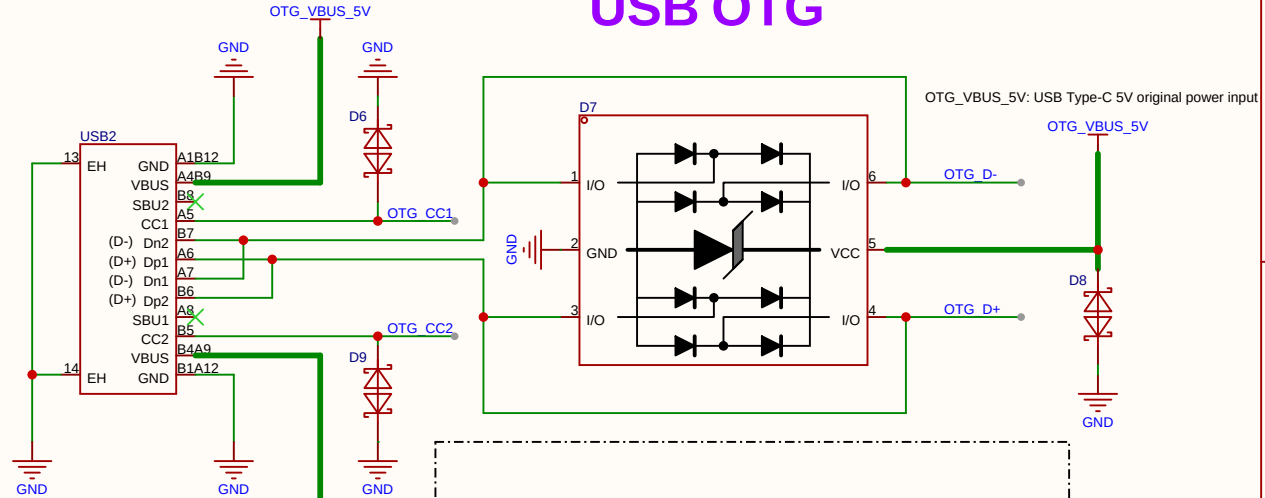
DTR	RTS	ESP_EN	ESP_BOOT	State
1	1	1	1	running
0	0	1	1	Safe state (no reset, no download)
1	0	0	1	RESET ESP32
0	1	1	0	Enter download mode

USB UART (CH343P)



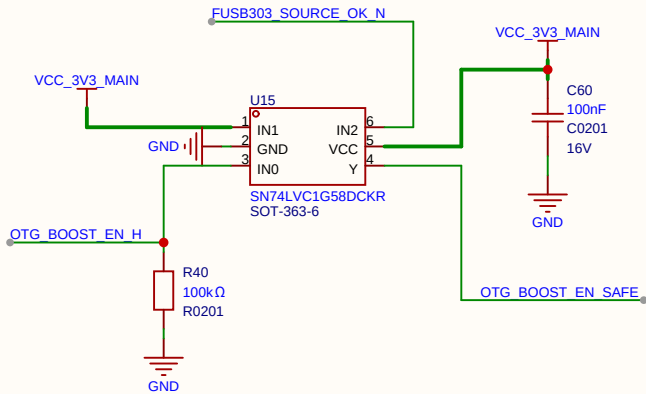


USB OTG

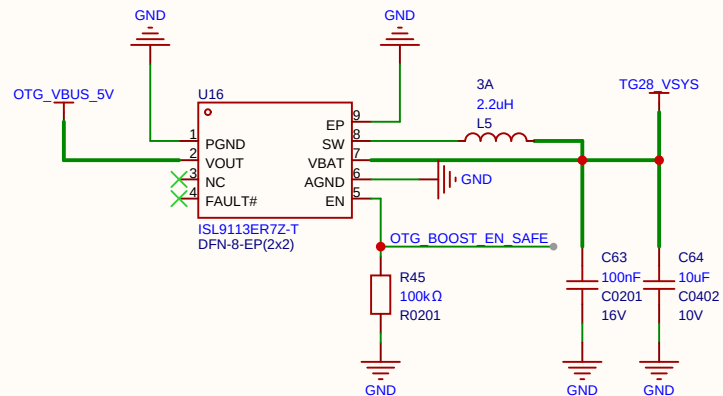


Note

Type-C2 VBUS is not connected to the system charging input.
 This interface does not support charging and dual-port power cabling.

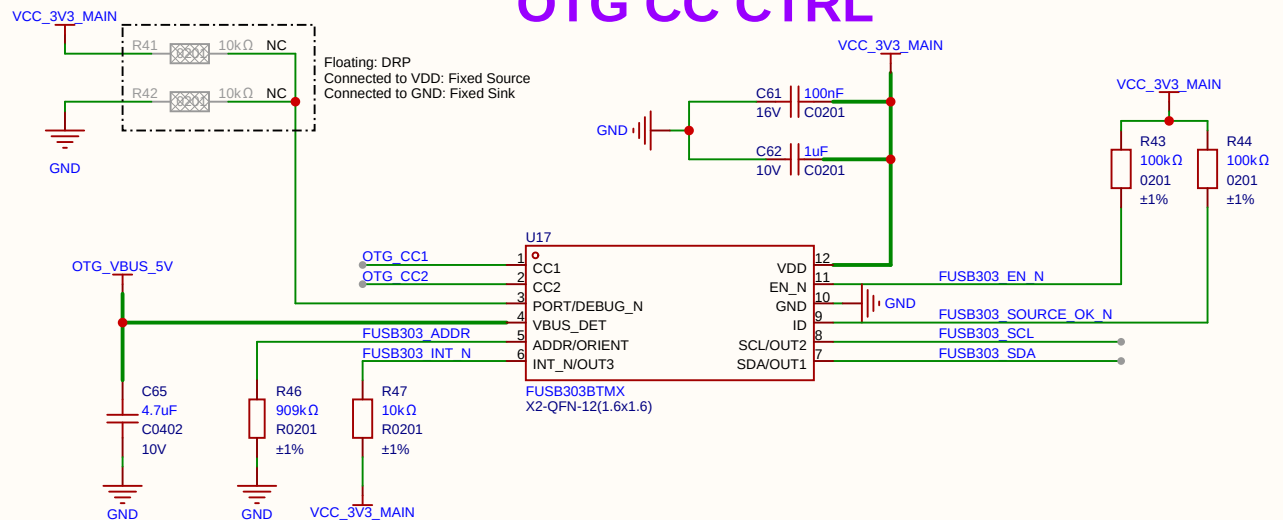


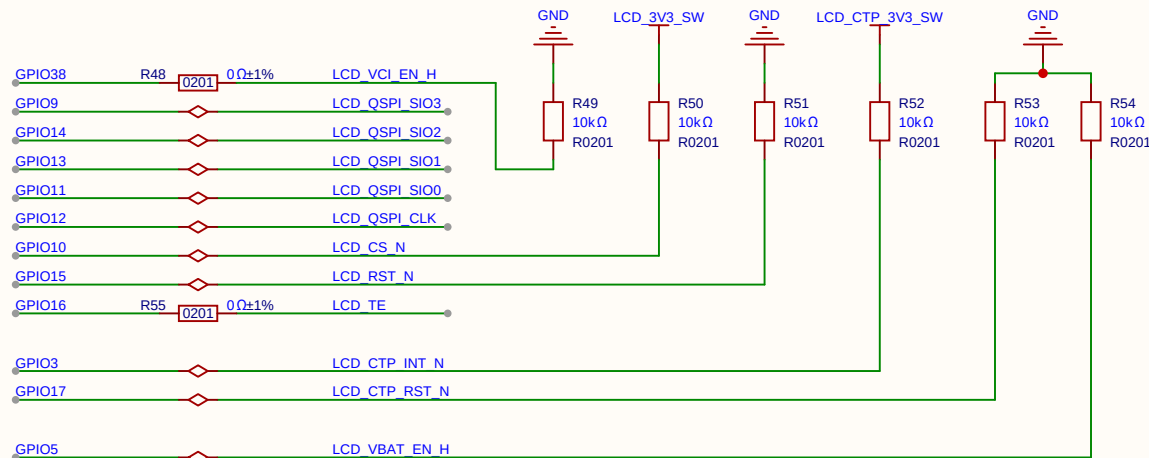
SOURCE_OK_N	GPIO	ISL9113
1	0	OFF
1	1	OFF (No Source)
0	0	OFF
0	1	ON (HOST 5V)



OTG BOOST

OTG CC CTRL



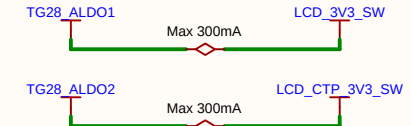


Note

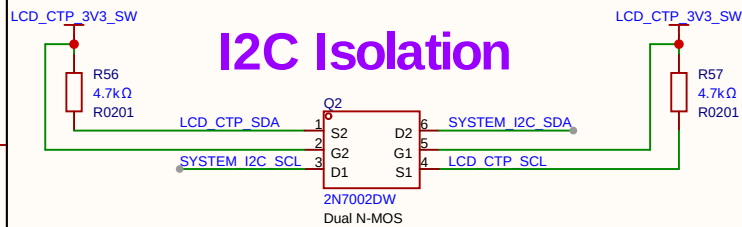
The power sequence must be followed to prevent direct current reverse injection!

Power Off:
 Set all relevant GPIO outputs to low or high impedance
 → Disable internal pull-ups
 → Wait for several microseconds
 → Shut down the camera/TF power supply

Turn on power:
 Turn on the power supply of the peripheral device
 → Wait for the power to stabilize
 → Then switch the GPIO multiplexing function



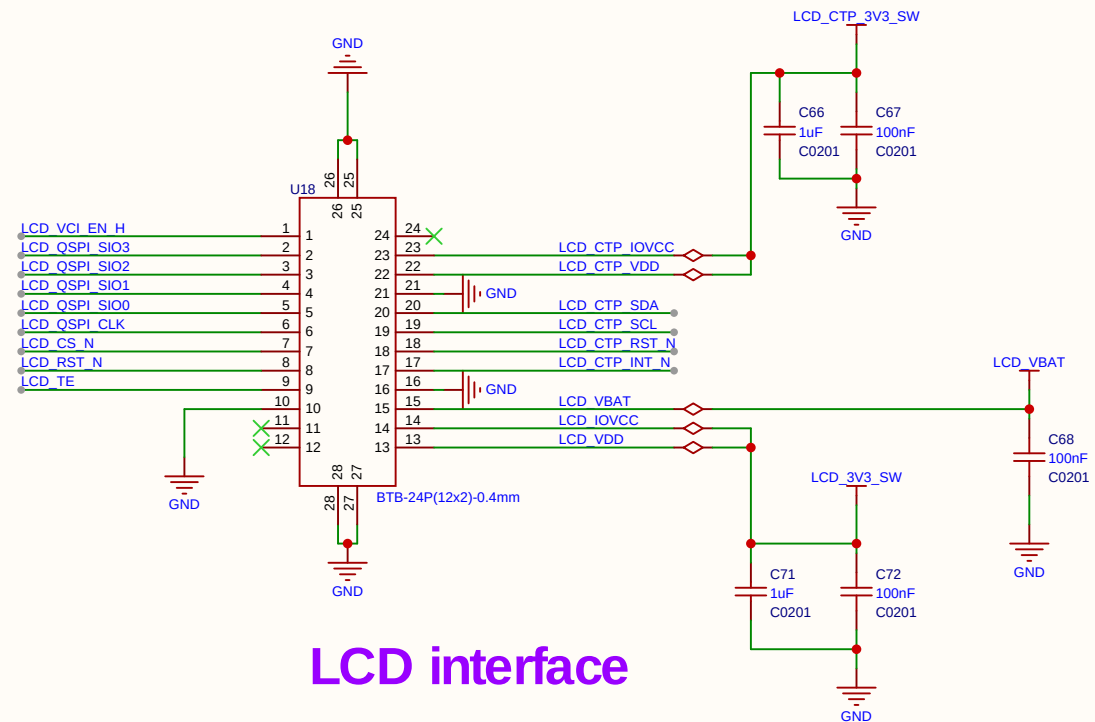
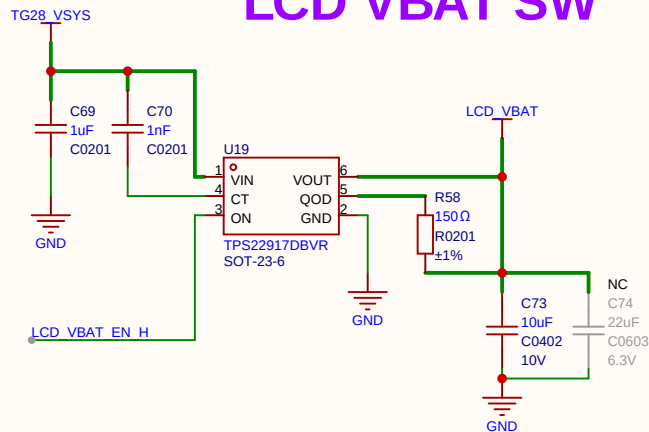
I2C Isolation



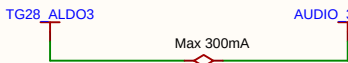
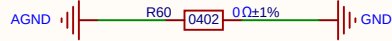
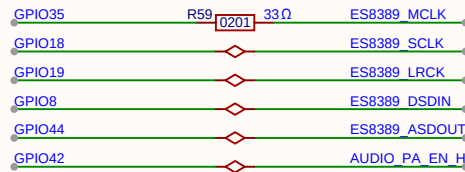
Note:
 CTP_SCL/CTP_SDA are isolated from SYSTEM_I2C by one dual N-MOS (2N7002DW).
 The SYSTEM_I2C side is pulled up to VCC_3V3_MAIN.
 The CTP side is pulled up to LCD_CTP_3V3_SW.
 Both MOS gates are tied to LCD_CTP_3V3_SW.

When LCD_CTP_3V3_SW is off, the MOSFETs are off and the body diodes are reverse-biased, so VCC_3V3_MAIN cannot back-power the unpowered CST820 touch controller through I2C.

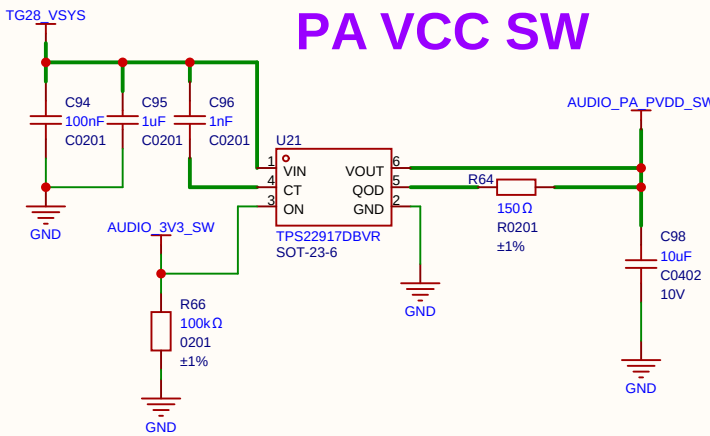
LCD VBAT SW



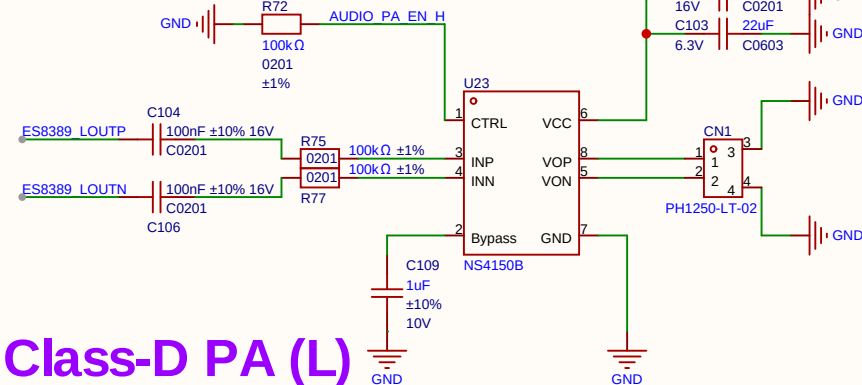
LCD interface



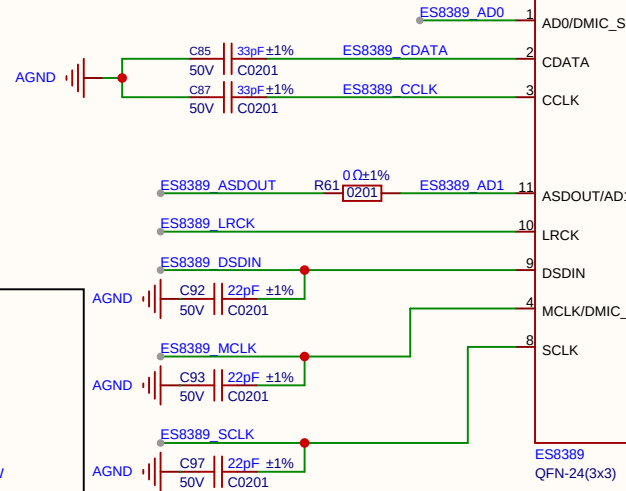
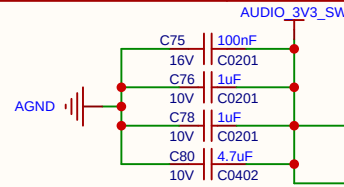
PA VCC SW



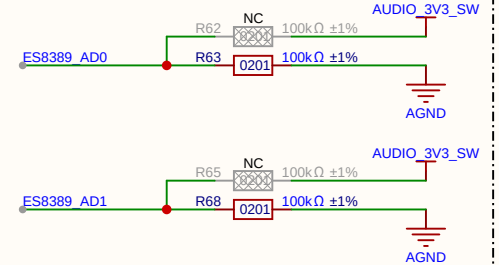
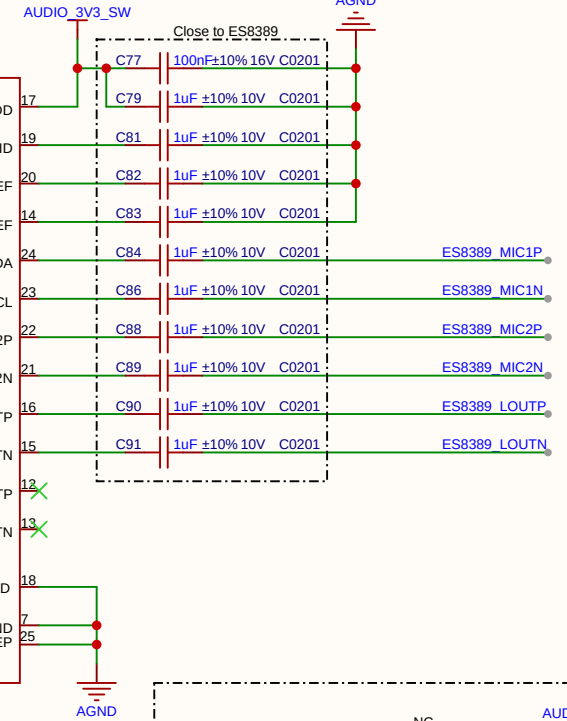
AUDIO_PA_PVDD_SW



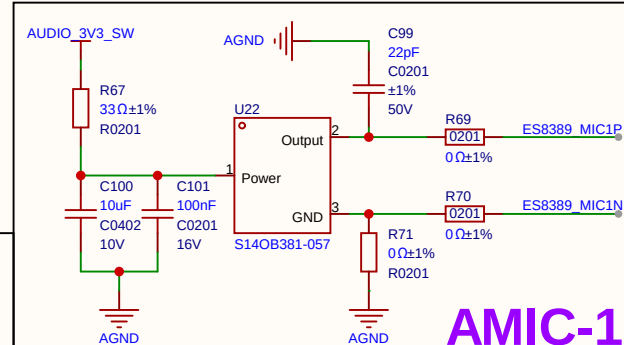
Class-D PA (L)



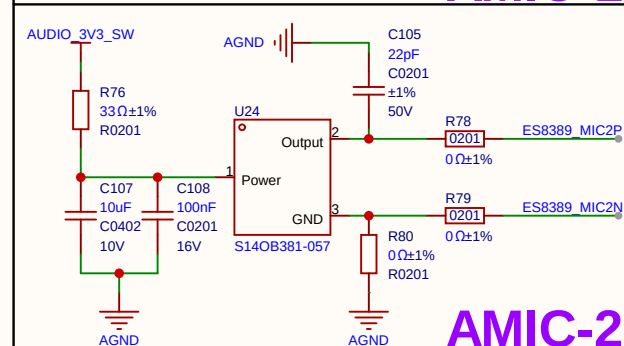
Codec



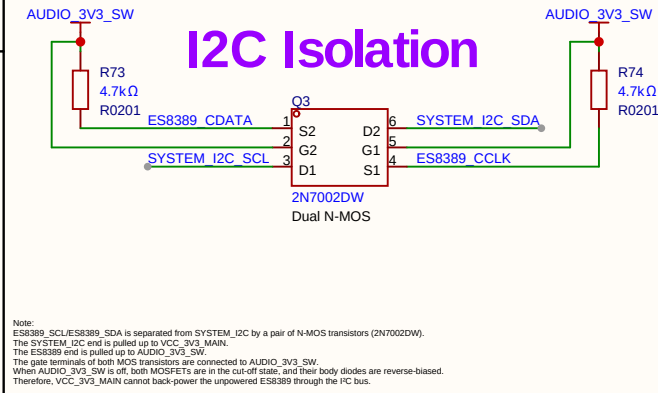
ADDR Config



AMIC-1



AMIC-2



Note:
ES8389_SCL/ES8389_SDA is separated from SYSTEM_I2C by a pair of N-MOS transistors (2N7002DW).
The SYSTEM_I2C end is pulled up to VCC_3V3_MAIN.
The ES8389 end is pulled up to AUDIO_3V3_SW.
When AUDIO_3V3_SW is off, both MOSFETs are in the cut-off state, and their body diodes are reverse-biased.
Therefore, VCC_3V3_MAIN cannot back-power the unpowered ES8389 through the I2C bus.

A

B

C

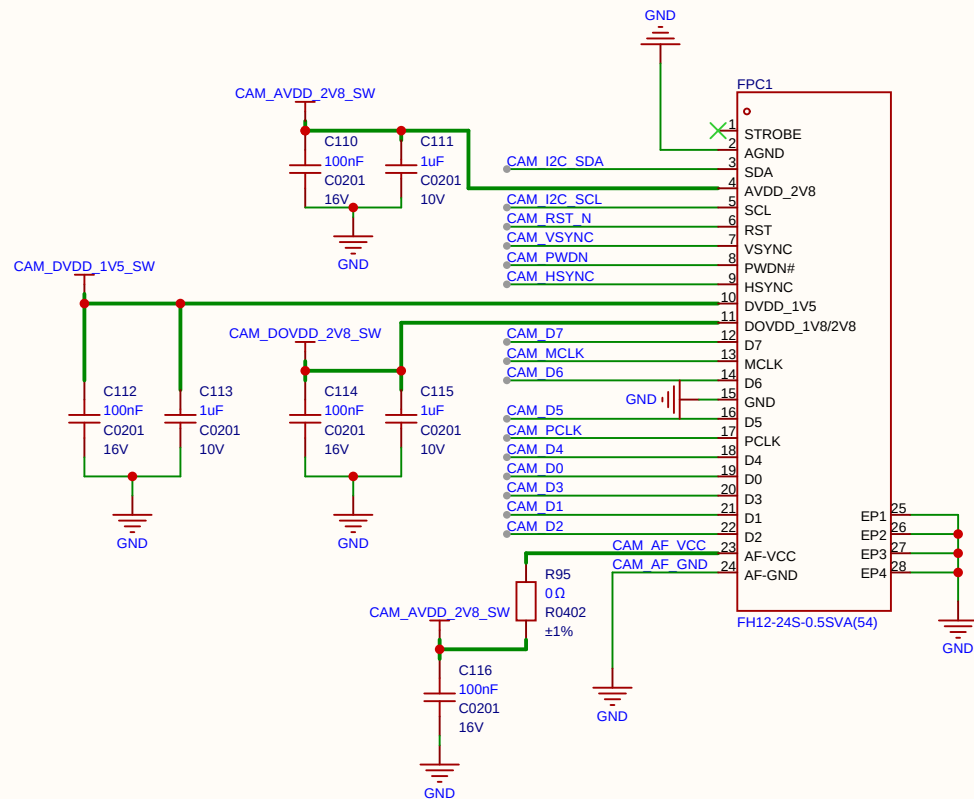
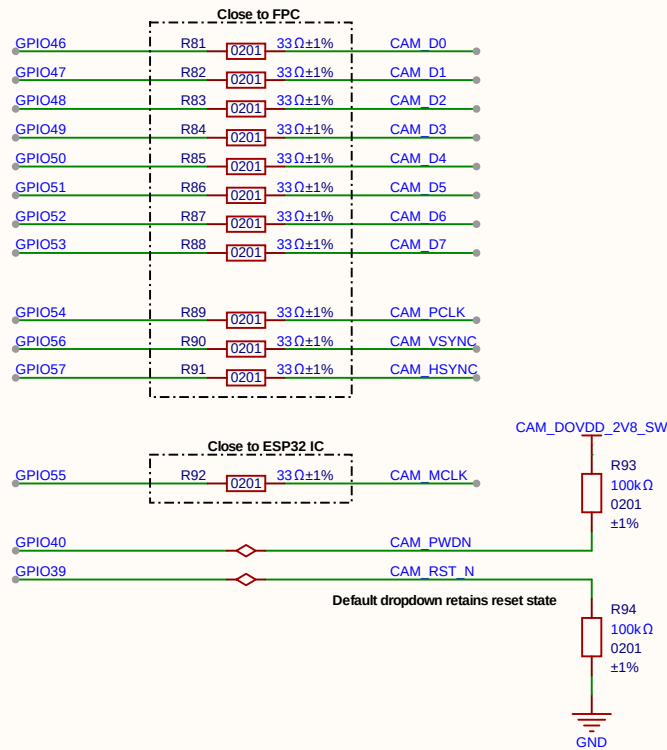
D

A

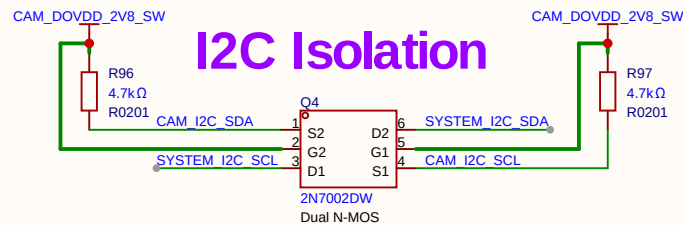
B

C

D



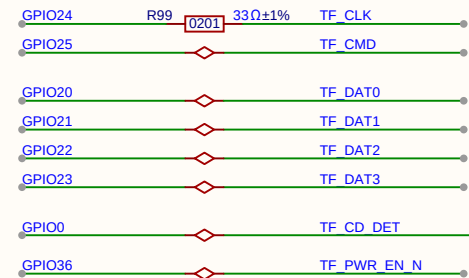
DVP Camera Interface



Note:
CAM_SCL/CAM_SDA is separated from SYSTEM_I2C by a pair of N-MOS transistors (2N7002DW).
The SYSTEM_I2C terminal is pulled up to VCC_3V3_MAIN.
The camera I2C is pulled up to CAM_DOVDD_2V8_SW.
The gates of the two MOS transistors are connected to CAM_DOVDD_2V8_SW.
When CAM_DOVDD_2V8_SW is turned off, the MOSFET is in the closed state and the body diode is in reverse bias.
Therefore, VCC_3V3_MAIN cannot provide a reverse power supply to an unpowered camera via I2C.

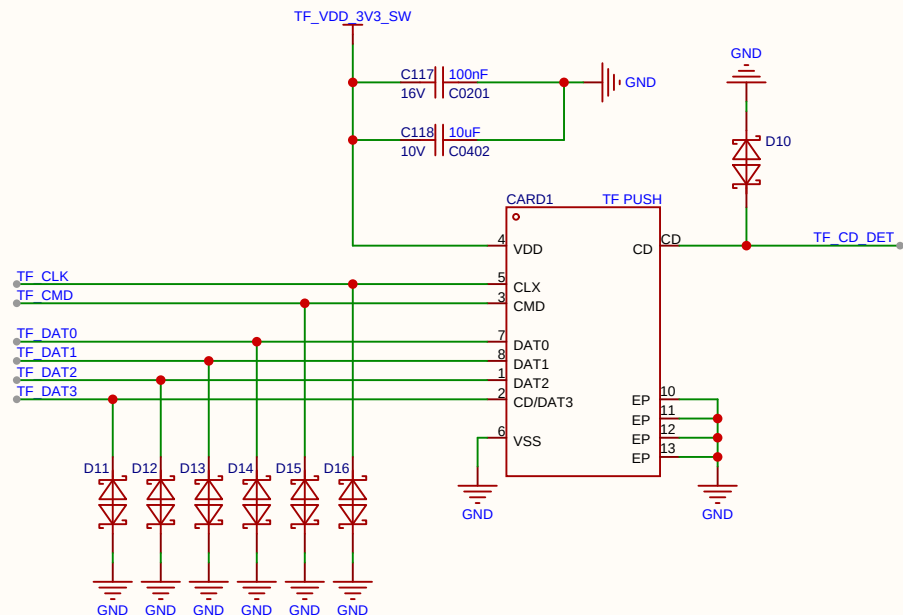
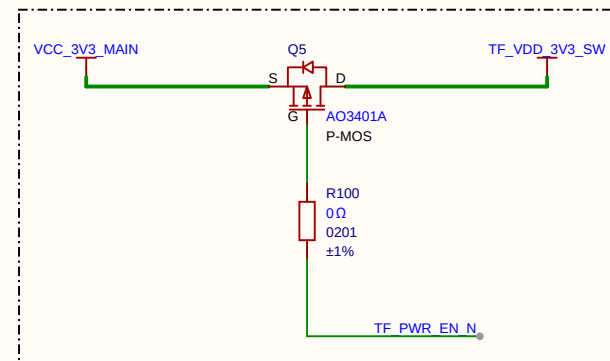
In the hardware design guide, it is clearly stated:

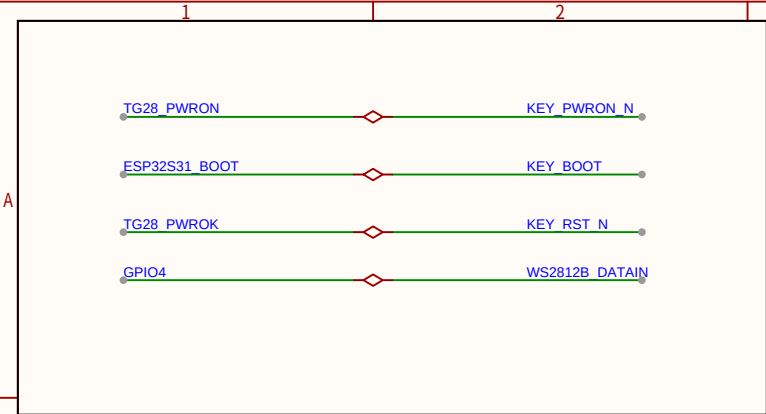
"When using slot0, since the GPIO power domain is an internal power supply, no pull-up resistor needs to be added."



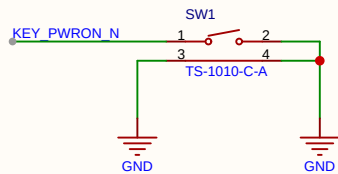
VCC_3V3_MAIN

R98
100kΩ
0201
±1%

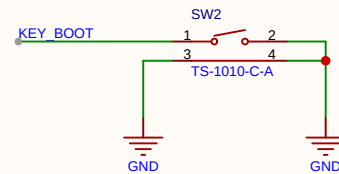




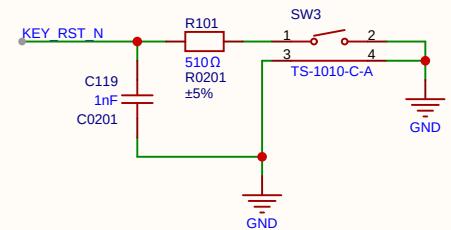
PWRON KEY



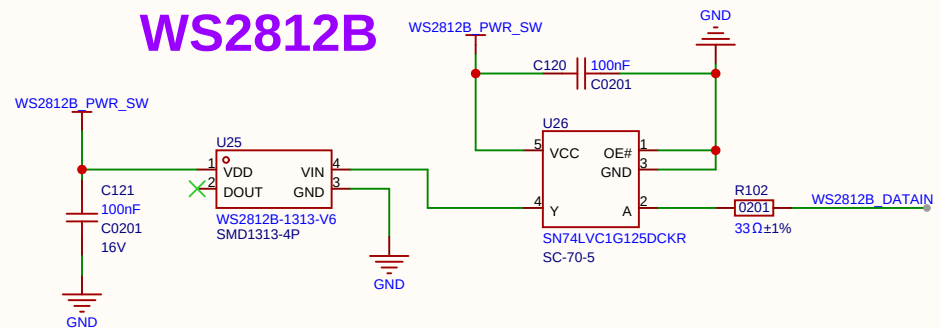
BOOT KEY



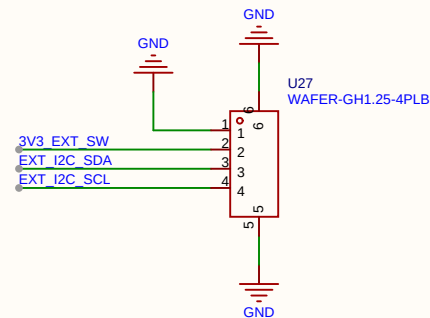
RESET KEY



WS2812B



GH1.25 4Pin EXT



I2C Isolation

